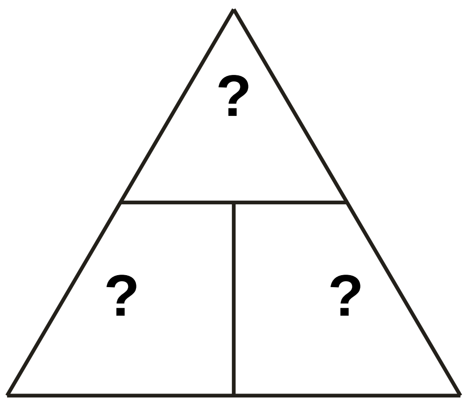
Cells: Structure, Transport & Division

Test · 33 marks total

Name: ________________________________	Date: ______________	Score: _____ / 33

Section A — Complete the formula triangle

A1. Fill in the three blanks with I, M and R in the correct places.



[3 marks]

Section B — Straight substitution

B1. A cell has an image size of 6 mm and a real size of 0.3 mm. Calculate the magnification.

[2 marks]

B2. A structure has an image size of 15 mm and a real size of 0.05 mm. Calculate the magnification.

[2 marks]

Section C — Rearrange and solve

C1. A real structure is 0.002 mm across and is magnified ×500. Calculate the image size, in mm. Show how you rearranged the equation.

[3 marks]

C2. An image measures 4 mm across at a magnification of ×800. Calculate the real size, in mm. Show how you rearranged the equation.

[3 marks]

C3. A real structure is 20 µm across and is magnified ×50. Calculate the image size, in mm. (Convert units first.)

[3 marks]

Section D — Cell structures

D1. Name two structures found in a plant cell but not in an animal cell.
[2 marks]

D2. Name two structures found in a bacterial cell that are not found in an animal cell.
[2 marks]

D3. State the function of the mitochondria.
[1 mark]

D4. State the function of a chloroplast.
[1 mark]

Section E — Transport across membranes

E1. Define diffusion.
[2 marks]

E2. Define osmosis, and state what makes it different from diffusion of a dissolved substance.
[2 marks]

E3. Explain why active transport requires energy from respiration, using the example of mineral ion uptake by plant root hairs.
[2 marks]

Section F — Cell division & differentiation

F1. State two reasons why body cells divide by mitosis.
[2 marks]

F2. State the number of chromosomes in a normal human body cell and in a human gamete, and explain why the gamete has this number.
[3 marks]